Homework 5: Adaptive Non-stiff ODE Solver

Problem Formulation

In this homework assignment, we aim to develop a solver for non-stiff ordinary differential equations. That is, ODEs of the form

$$\mathbf{y}' = \mathbf{f}(t, \mathbf{y}) \tag{1}$$

that satisfy $h\lambda \gg -\infty$.

I will use second-order Adams linear multistep methods, utilizing P(EC)E to reduce computational cost of implicit Adams-Moulton with Adams-Bashforth as the predictor method. I will also use adaptive step size control to obtain a numerical solution to (1) within a target error tolerance. Finally, I will use an interpolating polynomial to approximate the numerical solution at specific time points.

Numerical Schemes

A Linear Multistep Method is a numerical method which involves a linear combination of multiple steps to approximate the solution at the next time step. For non-constant step size Adams methods, we simply integrate (1) across a single timestep of size h_n

$$\mathbf{y}_n = \mathbf{y}_{n-1} + \int_{t_{n-1}}^{t_n} \mathbf{f}(t, \mathbf{y}(t))\, dt$$

The Adams methods approximate this integral using a polynomial interpolation of $\mathbf{f}(t, \mathbf{y}(t))$ using the past values of $\mathbf{f}$. One option for the interpolating polynomial is the Newton interpolation. The Newton interpolating polynomial passing through a set of points at times t_i with the coefficients computed by the divided difference table, defined recursively by

$$\mathbf{f}[t_n, t_{n-k}] = \frac{\mathbf{f}[t_n, t_{n-k+1}] - \mathbf{f}[t_{n-1}, t_{n-k}]}{t_n - t_{n-k}} \tag{2}$$

where $\mathbf{f}[t_n] = \mathbf{f}(t_n, \mathbf{y}(t_n))$. The polynomial is computed with

$$\phi(t) = \mathbf{f}[t_n] + \mathbf{f}[t_n, t_{n-1}](t - t_n) + \mathbf{f}[t_n, t_{n-2}](t - t_n)(t - t_{n-1}) + \cdots + \mathbf{f}[t_n, t_{n-k}](t - t_n) \cdots (t - t_{n-k+1}) \tag{3}$$

For second order Adams-Bashforth, we interpolate the past two $\mathbf{f}$'s. Integrating, we have the Adams-Bashforth method for non-constant step size

$$\mathbf{y}_n = \mathbf{y}_{n-1} + h_n \mathbf{f}_{n-1} \qquad \text{order 1} \tag{4}$$

$$\mathbf{y}_n = \mathbf{y}_{n-1} + h_n \mathbf{f}_{n-1} + \frac{h_n^2}{2h_{n-1}}(\mathbf{f}_{n-1} - \mathbf{f}_{n-2}) \qquad \text{order 2} \tag{5}$$

For Adams-Moulton, we interpolate across the current $\mathbf{f}$ first, and then the past $\mathbf{f}$'s. Thus, the Adams-Moulton method is given by the implicit scheme

$$\mathbf{y}_n = \mathbf{y}_{n-1} + h_n \mathbf{f}_n \qquad \text{order 1} \tag{6}$$

$$\mathbf{y}_n = \mathbf{y}_{n-1} + \frac{h_n}{2}(\mathbf{f}_n + \mathbf{f}_{n-1}) \qquad \text{order 2} \tag{7}$$

In order to avoid the computational cost and complexity of implementation introduced by the implicit Adams-Moulton method, we instead use the Adams-Bashforth method to predict $\mathbf{y}_n$ that is required by $\mathbf{f}_n$ in the Adams-Moulton method. The Predictor-Evaluate-Correct-Evaluate method first predicts $\mathbf{y}_n \approx \mathbf{y}_n^{(0)}$, which is only a good approximation in the non-stiff limit. $\mathbf{f}(t, \mathbf{y})$ is then evaluated at $(t_n, \mathbf{y}_n^0)$. The Corrector method uses this approximation to determine $\mathbf{y}_n$, after which $\mathbf{f}(\mathbf{t}, \mathbf{y})$ is evaluated at $(t_n, \mathbf{y}_n)$. The second-order Predictor-Corrector algorithm is as follows:

$$\begin{aligned}
&\text{Predict} \quad \mathbf{y}_n^{(0)} = \mathbf{y}_{n-1} + h_n \mathbf{f}_{n-1} + \frac{h_n^2}{2h_{n-1}}(\mathbf{f}_{n-1} - \mathbf{f}_{n-2}) \\
&\text{Evaluate} \quad \mathbf{f}_n^{(0)} = \mathbf{f}(t_n, \mathbf{y}_n^{(0)}) \\
&\text{Correct} \quad \mathbf{y}_n = \mathbf{y}_{n-1} + \frac{h_n}{2}(\mathbf{f}_n^{(0)} + \mathbf{f}_{n-1}) \\
&\text{Evaluate} \quad \mathbf{f}_n = \mathbf{f}(t_n, \mathbf{y}_n)
\end{aligned} \tag{8}$$

However, on the first time step, we only have the point $(t_0, \mathbf{y}(t_0))$, so we cannot use a two-step Adams-Bashforth method. In this case, we determine the starting values by using the one-step Adams-Bashforth method as the predictor and the two-step Adams-Moulton method as the corrector.

$$\begin{aligned}
&\text{Predict} \quad \mathbf{y}_n^{(0)} = \mathbf{y}_{n-1} + h_n \mathbf{f}_{n-1} \\
&\text{Evaluate} \quad \mathbf{f}_n^{(0)} = \mathbf{f}(t_n, \mathbf{y}_n^{(0)}) \\
&\text{Correct} \quad \mathbf{y}_n = \mathbf{y}_{n-1} + \frac{h_n}{2}(\mathbf{f}_n^{(0)} + \mathbf{f}_{n-1}) \\
&\text{Evaluate} \quad \mathbf{f}_n = \mathbf{f}(t_n, \mathbf{y}_n)
\end{aligned} \tag{9}$$

For constant step size, the previous equations are still used, but with the additional fact that $h_0 = h_1 = \cdots = h_n$.

Step Size Control

Evaluating the Solution at Specific Times

The Predictor-Corrector local truncation error is given by pg 150

Examples

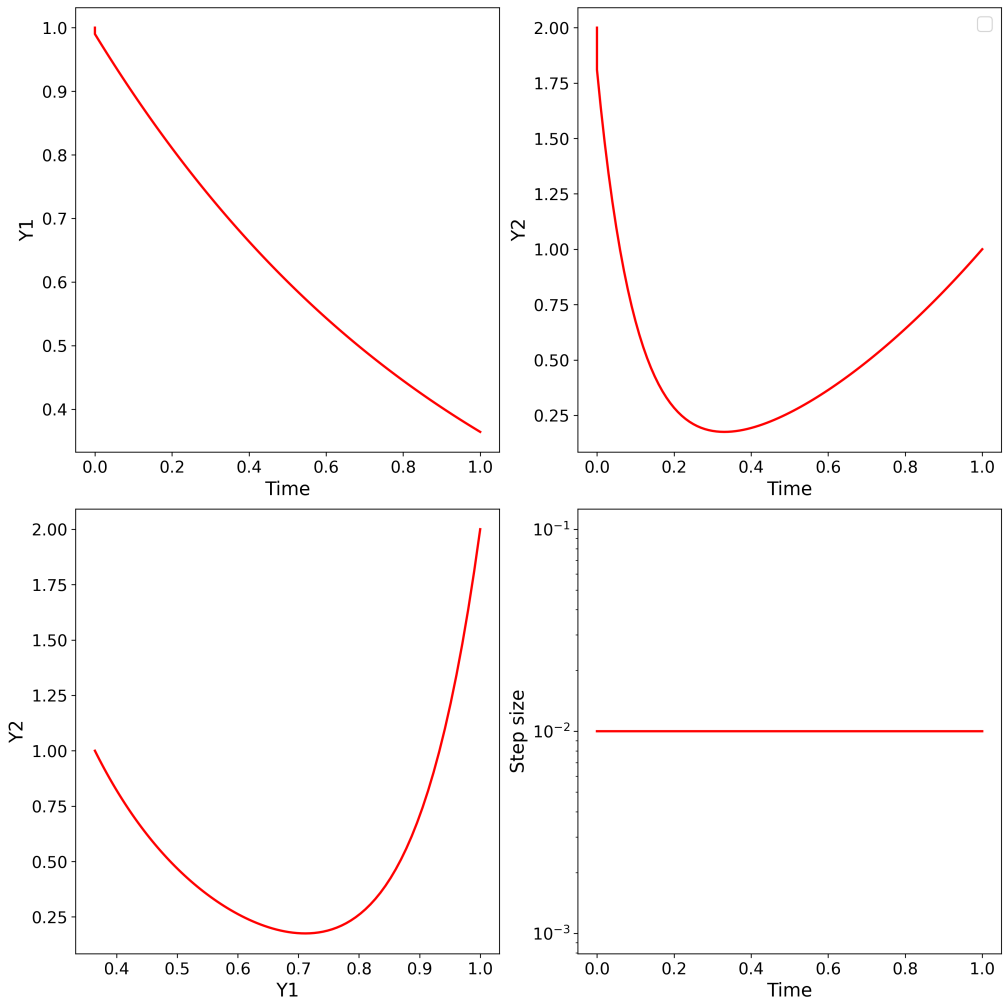

Figure 1: Constant step size

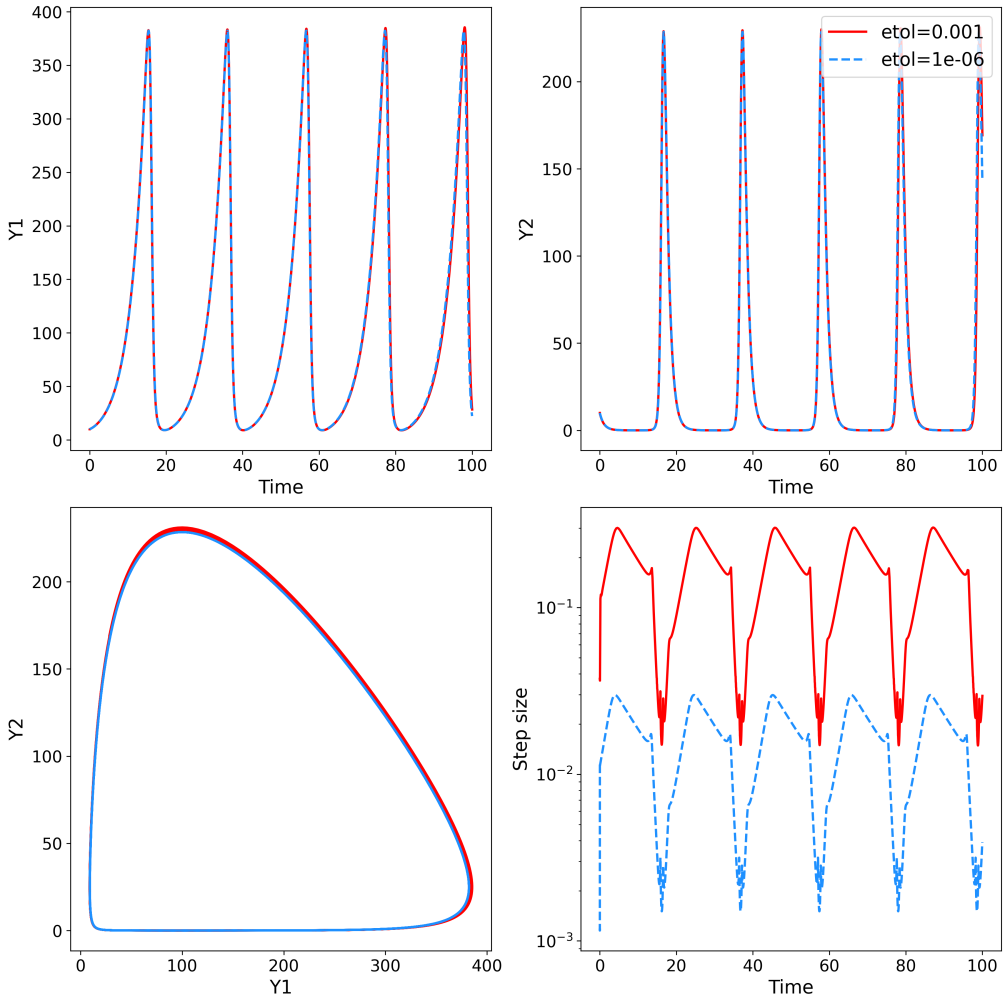

Figure 2: Predator-Prey problem

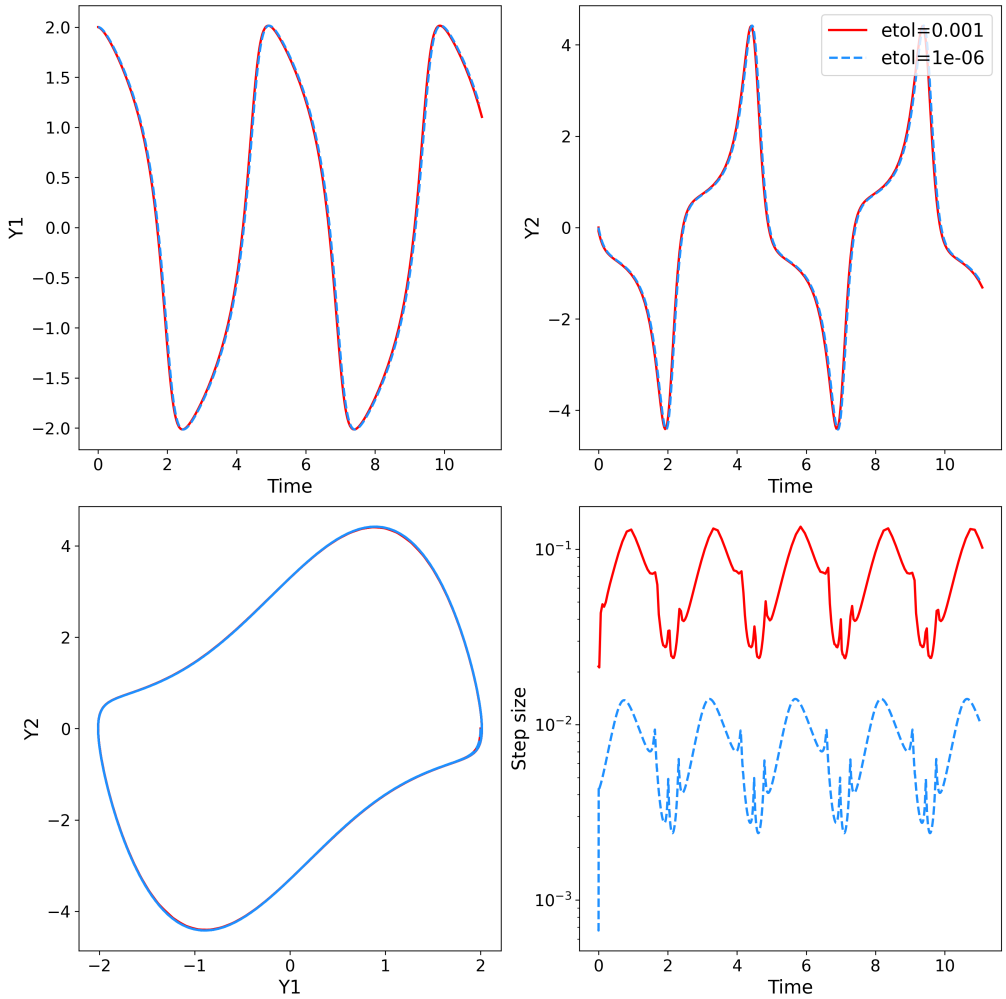

Figure 3: Van der Pol Oscillator

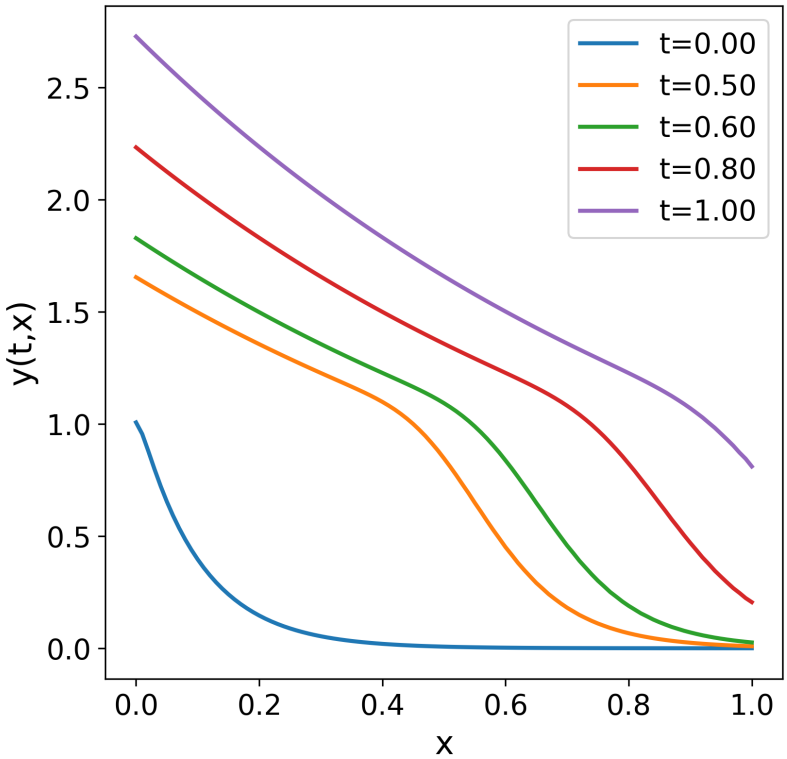

Figure 4: Method of Lines